

Transactions, Concurrency Control & Recovery - Detailed Notes

DETAILED COMPREHENSIVE NOTES

Subject: Database Management Systems | Chapter: Chapter 3: Transaction Processing, Concurrency Control & Recovery | Academic Year: 2023-2024

1. Transactions & The ACID Properties

A transaction is a logical unit of database processing that includes one or more database access operations. To maintain database consistency, transactions must adhere to ACID properties:

Atomicity: "All or nothing" execution guaranteed by the Transaction Manager and recovery system.

Consistency: Preserves database integrity constraints from one valid state to another.

Isolation: Intermediate transaction states are hidden from concurrently executing transactions, managed by Concurrency Control.

Durability: Committed updates persist permanently even in the event of system crashes or power failures.

2. Serializability & Concurrency Control Anomalies

Concurrent transaction execution without isolation introduces anomalies: Dirty Reads (reading uncommitted data), Non-repeatable Reads (different reads within same transaction), and Phantom Reads (new rows inserted during range queries).

Conflict Serializability: A schedule is conflict serializable if it can be transformed into a serial schedule by swapping non-conflicting adjacent operations. Tested via Precedence Graph (Serialization Graph) where cycles denote non-serializable schedules.

3. Locking Protocols & Database Recovery

Two-Phase Locking (2PL): Growing Phase (acquiring locks, releasing none) followed by Shrinking Phase (releasing locks, acquiring none). Guarantees conflict serializability.

Strict 2PL: Holds all exclusive locks until commit/abort, preventing cascading rollbacks.

Recovery techniques utilize Write-Ahead Logging (WAL): Log records must be flushed to non-volatile storage before corresponding database disk blocks are written.

ARIES Algorithm implements: Analysis phase (identifies active transactions and dirty pages), Redo phase (repeats history to reconstruct state), and Undo phase (rolls back active uncommitted transactions).